

Stepping Through The Looking Glass, WE FOUND DISPLAYS SO DIFFERENT



Janani Gopalakrishnan Vikram is a technically-qualified freelance writer, editor and hands-on mom based in Chennai

Eyes, perfected by Mother Nature, are matchless display systems. Every other man-made display has some problem or the other—battery life, heat, glare, image quality and what not. However, humans never give up on their desire to out-do or, at least, match nature. And so, manufacturers and researchers continue attempting to improve the current display systems, making these smaller, more flexible, power-efficient, panoramic, etc.

Moreover, ubiquity of devices has also led to the need for displays that are flexible and have a smaller footprint. Mobiles, wearables, augmented reality (AR) workspaces, immersive games and other applications throw challenges at the display industry to match their speed of travel into the future. What will the future displays look like? In this story, we dig into a few off-beat and futuristic research projects that give us a few clues regarding what displays might be like a decade from now.

Ole! Time to bend, roll and tango

Organic light emitting diodes (OLEDs) have been around for quite some time now. Having started their career with mobile devices, these are now beginning to be used in high-end large-size televisions, too. Made of organic materials that glow when subjected to electric current, these screens have laudable energy savings, resolution and form factor. More attractively, these organic materials are more flexible and make flexible displays a reachable goal.

Manufacturers like LG and Samsung are especially keen on bendy displays. A few years ago, Samsung demonstrated YOUM, a flexible active-matrix OLED (AMOLED) display that uses polyimide as the substrate material instead of glass. Polyimide is a polymer with high resilience

and flexibility. Although Samsung has used this technology in a few of its curved phone screens, the impact has not been felt much because most of these displays have been encased in toughened Gorilla glass.

However, it is clear that the company is keen on flexible displays and hopes to use it to make jaw-dropping devices. A patent filed by Samsung and published by the US Patent and Trademark Office (USPTO) last November (<http://tinyurl.com/h9wpbtq>) focuses on such scrollable, foldable smartphones and tablets. It describes a device that can be folded in half like a book, and another that can be rolled up into a cylinder.

The patent also hints at what the foldable device could possibly do in open and closed states. When in closed state during a call, the device would show the face of the caller and call-related options like recording, keypad and mute, while the open state will additionally have the contact list opened in the other half of the display. Other applications like messaging, images, etc, are also shown using the flexible display. Some images also suggest notifications being shown on the back of the screen when folded on the sides, or the corner.

LG is equally keen on flexible display tech. They showed off some beautiful tech at this year's Consumer Electronics Show (CES 2016). Apart from a 45.7cm (18-inch) flexible display panel, they showed two smaller versions, one that was transparent and another that could be rolled up into a 3cm-radius cylinder.

LG claims that their new OLED displays are flexible enough to curve around walls or be used seamlessly inside cars. These are also thinner, lighter and break-resistant; so light that a thin magnetic strip is all that is needed to stick the 4mm-thick television screen to the wall and so convenient to peel off and stick somewhere else, as required.

The display produces its own light, so

Conceptual smartphones with transparent and flexible display

